

ANTI COLLISION SPEED CONTROL SYSTEM BASED ON RFID

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Abstract

Road accidents have become one of the major causes of death and injuries throughout the world. Increasing vehicle density, over speeding, careless driving, poor visibility, and lack of proper collision prevention systems are some of the major reasons for traffic accidents. In industrial areas such as mines and warehouses, collisions between robotic vehicles and moving equipment also create serious safety risks and operational losses. Therefore, there is a growing need for intelligent safety systems that can detect possible collisions in advance and automatically control vehicle speed to avoid accidents.

The proposed “Anti-Collision Speed Control System Based on Embedded System Using RFID” is designed to provide an effective and reliable solution for preventing collisions and improving safety in transportation and industrial environments. The proposed system utilizes Radio Frequency Identification (RFID) technology along with an embedded microcontroller to detect nearby vehicles, robots, turns, and hazardous zones. Passive High-Frequency RFID tags are placed at specific locations such as sharp turns, steep gradients, restricted areas, and collision-prone zones. RFID readers mounted on the vehicle continuously scan for these tags while the vehicle is in motion. Whenever an RFID tag is detected, the information is sent to the microcontroller, which processes the data and takes necessary action to control the speed of the vehicle automatically.

Keywords

RFID Technology, Anti-Collision System, Speed Control, Embedded System, Road Safety, Vehicle Automation, Accident Prevention, Microcontroller, Intelligent Transportation System (ITS), RFID Reader, RFID Tags, Automatic Braking

1. Introduction

The Indian Law Commission has issued a recommendation to decrease traffic accidents, limit speed in sensitive areas and promote public tranquility. Due to some driver's rash driving, the current techniques are still unable to reduce accidents [1]- [2]. Thus, speed control must be implemented in all automobiles. Installing an automated speed control system in cars, notably in constrained areas, is our brandnew recommendation. A gadget is set up as a transmitter in this case[3].The microprocessor analyses the vehicle's current speed to the specified limit and automatically adjusts the speed after receiving the measurement from the dc motor. This system uses Zigbee technology, which has a range of up to 10 to 100 meters, to communicate between the transmitter and receiver. [3]-[5] Compared to other options, this is less expensive. As a result, this system controls and monitors every vehicle in the area it covers. In our quick-paced environment, accidents are decreased by using this approach. People in both developed and developing nations experience inconvenience from traffic accidents and vehicle jams caused by drivers who disobey the law in a restricted zone, where the speed must be limited according to the zone's

regulations. Zigbee technology is used in automated speed control systems to limit the speed automatically. [6]-[9] It is sufficient to mention at this point that the introduction of several highway safety measures, including speed limits, among others, has significantly lowered the frequency of these incidents. The problem is that safe driving policies alone won't solve this; engineers must also play a part because the root cause of the problem is an engineering product (the motor vehicle). Many drivers have been forced out of necessity to make the trip through dark areas while also being quite fatigued. Doing this isn't necessarily a bad idea. Many examples that have been reported include drivers who fell asleep at the wheel; when they finally awoke, a head-on accident may have occurred. Few people have had the good luck to immediately stop this. Therefore, it is essential to take into account the benefits of an early warning system where the driver is informed of a potential collision a significant period of time in respective time interval [10] –[11].

2. Literature Review

1. A. Mohammed Irfan et al. The authors presented an RFID-based intelligent transportation safety system integrated with embedded controllers and warning devices. The system continuously monitored vehicle movement and generated alerts during dangerous conditions. Their research demonstrated that RFID technology provides reliable contactless communication and improves accident prevention efficiency.

2. T. Ganesh and M. Arun Kumar The authors developed an automated anti-collision and warning system using RFID, sensors, and embedded controllers. The system detected nearby vehicles and hazardous conditions, then activated alarms and speed control mechanisms to prevent accidents. Their work demonstrated that integrating RFID with embedded systems improves safety, reduces human error, and

enhances transportation reliability in industrial and road environments.

3. Ajith Kumar and D. Bharathi In their study on intelligent road safety systems, the authors implemented RFID-based accident prevention technology for smart transportation. RFID readers installed in vehicles communicated with roadside tags to identify danger zones and control speed automatically. The researchers concluded that the system provides reliable accident prevention and supports future smart city applications.

4. K. Balamurugan et al. The authors designed a mining vehicle safety system capable of detecting sharp turns, steep gradients, and nearby obstacles using RFID technology. The embedded controller automatically controlled vehicle speed based on environmental conditions. Their research emphasized that automated safety systems improve operational efficiency and worker safety in mining industries.

5. M. Praveen Kumar et al. In their research on “Automatic Speed Control System Using RFID Technology”, the authors designed a smart vehicle control system capable of reducing vehicle speed automatically near schools, hospitals, and sharp turns. RFID tags installed at critical zones transmitted information to the vehicle controller, which adjusted speed accordingly. The researchers concluded that the system helps reduce accidents caused by human negligence and overspeeding.

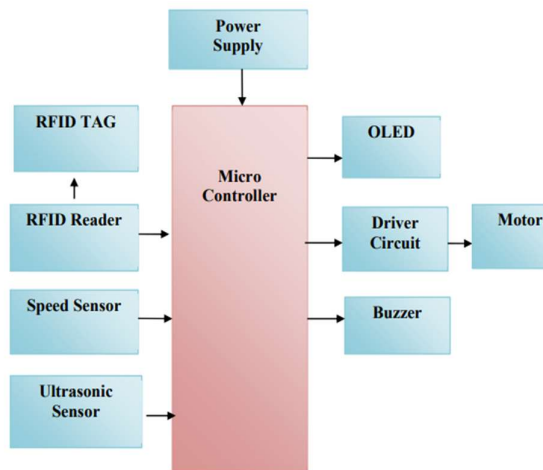
3. Existing System

The existing anti-collision systems in vehicles typically use sensors like cameras or radar to detect obstacles and alert drivers. However, these systems do not always control the vehicle's speed or directly prevent collisions in real-time.

4. Proposed System

The proposed anti-collision speed control system uses an Embedded System with RFID technology to automatically detect nearby vehicles or obstacles and adjust the speed of the vehicle accordingly, preventing collisions. The RFID tags on other vehicles communicate with the system to enable real-time speed adjustments, enhancing safety and reducing the risk of accidents.

5. Block Diagram



Working Principle :

The above block diagram represents the working principle of the “AntiCollision Speed Control System Based on Embedded System Using RFID.” The system is designed to detect obstacles, monitor vehicle movement, and automatically control the speed of the vehicle to prevent accidents. The complete system mainly consists of RFID technology, sensors, embedded controller, motor driver circuit, display unit, and alert mechanisms.

The power supply unit provides the necessary electrical power for all components in the system. The microcontroller acts as the central processing unit and controls the overall operation of the system. The RFID reader continuously scans for RFID tags placed at dangerous zones such as sharp turns, steep slopes, restricted areas, and collision-prone locations. When

the RFID reader detects a tag, the information is sent to the microcontroller for processing.

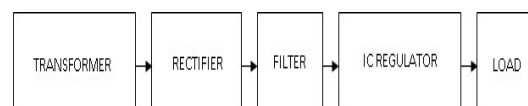
The speed sensor continuously monitors the speed of the vehicle and sends realtime data to the controller. The ultrasonic sensor is used to detect nearby obstacles or vehicles present in front of the system. Based on the information received from the RFID reader, speed sensor, and ultrasonic sensor, the microcontroller decides the appropriate speed control action required to avoid collisions.

Hardware Implementation :

- Power supply
- Bridge Rectifier
- Voltage Regulators
- Arduino UNO
- OLED
- Ultrasonic Sensor
- Transducers
- Speed Sensor
- RFID
- RFID Reader
- RFID Tag
- ULN2003
- Relay
- DC Motor
- Buzzer

Power Supply:

The power supply section is the section which provide +5V for the components to work. IC LM7805 is used for providing a constant power of +5V. The ac voltage, typically 220V, is connected to a transformer, which steps down that ac voltage down to the level of the desired dc output. This resulting dc voltage usually has some ripple or ac voltage variation.



Bridge Rectifier:

When four diodes are connected as shown in figure, the circuit is called as bridge rectifier. The input to the circuit is applied to the diagonally opposite corners of the network, and the output is taken from the remaining two corners.

Let us assume that the transformer is working properly and there is a positive potential, at point A and a negative potential at point B the positive potential at point A will forward bias D3 and reverse bias D4.

MICROCONTROLLER:

Arduino/genuino uno is a microcontroller board based on the atmega328p (datasheet). It has 14 digital input/output pins (of which 6 can be used as pwm outputs), 6 analog inputs, a 16 MHz quartz crystal, a usb connection, a power jack, an icsp header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a usb cable or power it with a ac-to-dc adapter or battery to get started.. You can tinker with your uno without worrying too much about doing something wrong, worst case scenario you can replace the chip for a few dollars and start over again

OLED Display:

An OLED display have the following advantages over an LCD display:

>Improved image quality - better contrast, higher brightness, fuller viewing angle, a wider colour range and much faster refresh rates.

>Lower power consumption.

>Simpler design that enables ultra-thin, flexible foldable and transparent displays.

Better durability - OLEDs are very durable and can operate in a broader temperature range.

ULTRASONIC SENSOR:

Ultrasonic sensors are industrial control devices that use sound waves above 20,000 Hz, beyond the range of human hearing, to measure and calculate distance from the sensor to a specified target object.

Transducers:

Sound field of a non focusing 4MHz ultrasonic transducer with a near field length of $N=67\text{mm}$ in water. The plot shows the sound pressure at a logarithmic db scale. Sound pressure field of the same ultrasonic transducer (4MHz, $N=67\text{mm}$) with the transducer surface having a spherical curvature with the curvature radius $R=30\text{mm}$ An ultrasonic transducer is a device that converts energy into ultrasound, or sound waves above the normal range of human hearing. While technically a dog whistle is an ultrasonic transducer that converts mechanical energy in the form of air pressure into ultrasonic sound waves, the term is more apt to be used to refer to piezoelectric transducers that convert electrical energy into sound. Piezoelectric crystals have the property of changing size when a voltage is applied, thus applying an alternating current (AC) across them causes them to oscillate at very high frequencies, thus producing very high frequency sound waves.

SPEED SENSOR:

A speed sensor is a type of tachometer. It is a sender device used for reading the speed of a vehicle's wheel rotation. It usually consists of a toothed ring and pickup.

Differential hall-effect speed sensors for reliable operation in harsh environment and high temperature applications. Custom-designed to your requirements.

RADIO FREQUENCY IDENTIFICATION:

RFID, short for Radio Frequency Identification, is a technology that enables identification of a tag (that is normally attached with an entity) by using electromagnetic waves. RFID Reader Module, are also called as interrogators. They convert radio

waves returned from the RFID tag into a form that can be passed on to Controllers, which can make use of it. RFID tags and readers have to be tuned to the same frequency in order to communicate. RFID systems use many different frequencies, the most common and widely used & supported by our Reader is 125 KHz.

RFID Reader:

The reader, or scanner, functions similarly to barcode scanner; however, while a barcode scanner uses a laser beam to scan the barcode, an RFID scanner uses electromagnetic waves. To transmit these waves, the scanner uses an antenna that transmits a signal, communicating with tags antenna. The tags antenna receives data from the scanner and transmits its particular chip DC information to the scanner.

RFID Tag:

RFID tag is a small device which stores and sends data to RFID reader. They are categorized in two types – active tag and passive tag. Active tags are those which contain an internal battery and do not require power from the reader. Typically active tags have a longer distance range than passive tags. Passive tags are smaller and lighter in size than the active tags. They do not contain an internal battery and thus depend on RFID reader for operating power and certainly have a low range limited up to few meters.

ULN2003:

The ULN2003 is a monolithic IC consists of seven NPN darlington transistor pairs with high voltage and current capability. It is commonly used for applications such as relay drivers, motor, display drivers, led lamp drivers, logic buffers, line drivers, hammer drivers and other high voltage current applications. It consists of common or cathode clamp diodes for each NPN darlington pair which makes this driver IC useful for switching inductive loads.

RELAY:

A relay is an electromechanical switch, which perform ON and OFF operations without any human interaction. General representation of but double contact relay is shown in fig. Relays used where it is necessary to control a circuit by a lowpower signal (with complete electrical isolation between control and controlled circuits), or where several circuits must be controlled by one a signal.

DC MOTOR:

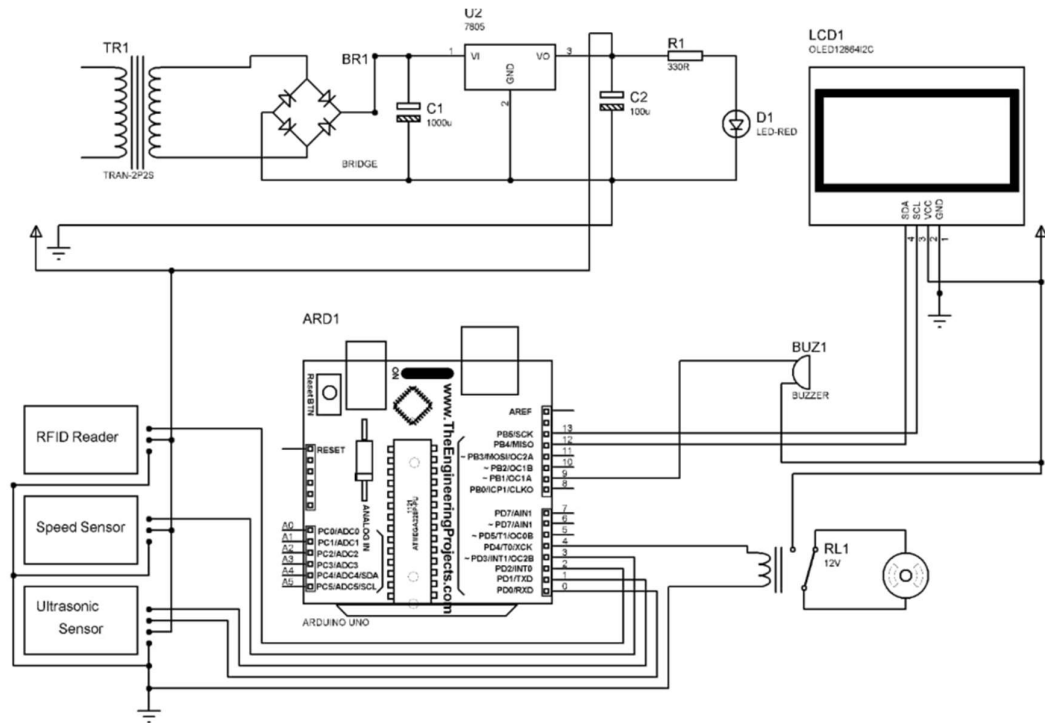
A DC motor in simple words is a device that converts direct current(electrical energy) into the mechanical energy. It's of vital importance for the industry today. A DC motor is designed to run on electric power. Two examples of pure DC designs are Michael Faraday's homo-polar motor (which is uncommon), and the ball bearing motor, which is (so far) a novelty. By far the most common DC motor types are the brushed and brushless types, which 37 use internal and external commutation respectively to create an oscillating AC current from the DC source—so they are not purely DC machines in a strict sense.

BUZZER:

A buzzer or beeper is a signaling device, usually electronic, typically used in automobiles, house hold appliances such as a microwave oven, or game shows.

It most commonly consists of a number of switches or sensors connected to a control unit that determines if and which button was pushed or a preset time has lapsed, and usually illuminates a light on the appropriate button or control panel, and sounds a warning in the form of a continuous intermittent buzzing or beeping sound. Initially this device was based on an electromechanical system which was identical to an electric bell without the metal gong (which makes the ringing noise).

SCHEMATIC DIAGRAM :



RESULTS :

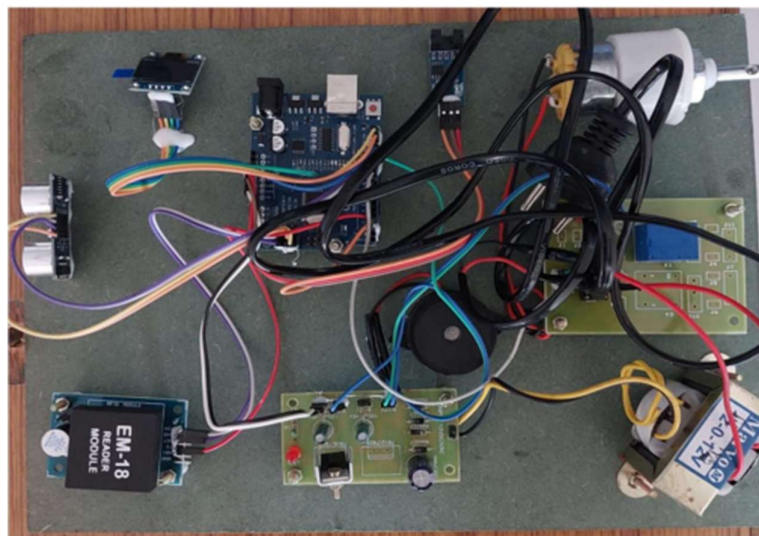
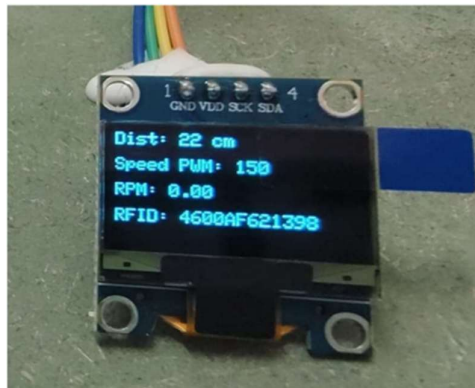


Fig: Implementation

OUTPUT:



CONCLUSION:

The proposed “Anti-Collision Speed Control System Based on Embedded System Using RFID” provides an effective solution for improving road safety and reducing accidents caused by over speeding and human negligence. The system automatically detects hazardous zones, nearby vehicles, and obstacles using RFID technology and ultrasonic sensors, and accordingly controls the speed of the vehicle. This automatic speed regulation helps drivers maintain safe driving conditions, especially in restricted areas, sharp turns, steep terrains, and mining fields where accidents are more likely to occur. The embedded system continuously monitors sensor inputs and takes quick decisions to reduce vehicle speed whenever dangerous situations are detected. The use of RFID technology improves detection accuracy and response time, while the ultrasonic sensor enhances obstacle detection capability. The buzzer and OLED display provide warning indications to alert the driver about possible danger conditions. Thus, the system minimizes human error and improves overall transportation safety. One of the major advantages of the proposed system is its reliability, low cost, and easy implementation. The system can be widely used in automobiles, industrial robots,

mining vehicles, and warehouse transportation systems. Future enhancements such as IoT integration, GPS tracking, wireless communication, and advanced sensor technologies can further improve the performance and intelligence of the system. Therefore, the proposed project offers a smart, reliable, and efficient solution for modern vehicle safety and collision prevention applications.

FUTURE SCOPE :

1. Integration with GPS Technology Future systems can combine RFID with GPS to provide real-time vehicle tracking and location-based speed control.
2. Automatic Braking System The project can be upgraded to automatically apply brakes when a collision risk or restricted speed zone is detected.
3. IoT-Based Vehicle Monitoring Using IoT technology, vehicle data such as speed, location, and driver behavior can be monitored remotely through cloud platforms.
4. Smart Traffic Signal Communication Vehicles can communicate directly with smart traffic signals to receive speed instructions and traffic updates automatically.

5. AI-Based Collision Prediction Artificial Intelligence and Machine Learning algorithms can be used to predict possible accidents based on road conditions and vehicle movement.

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